

CHAPTER 1

INTRODUCTION

Waste management is one of the major environmental challenges faced by modern society. The increasing amount of waste generated in homes, industries, and public places has made proper waste segregation essential for effective recycling and disposal. Traditional waste management methods rely heavily on manual segregation, which is time-consuming, labour-intensive, and often inaccurate. Improper segregation can lead to environmental pollution, health hazards, and reduced recycling efficiency.

The IoT-Based Smart Dustbin Segregation System is developed to automate the process of waste segregation using sensors, a microcontroller, and IoT technology. The system detects the type of waste and automatically directs it to the appropriate compartment, reducing human effort and improving accuracy. Additionally, IoT connectivity enables real-time monitoring of dustbin status, making waste management more efficient and intelligent. This system can be implemented in homes, schools, offices, and smart cities to promote a cleaner and more sustainable environment.

1.1 Motivation

Improper waste disposal has become a major concern worldwide. Many recyclable materials are mixed with general waste, reducing recycling efficiency and increasing the burden on landfills. Manual segregation methods are often inefficient and expose workers to health risks.

The motivation behind developing this project is to create an intelligent and automated waste segregation system that minimizes human intervention while improving waste management efficiency. By using IoT technology and smart sensors, the system can automatically classify waste and provide real-time monitoring capabilities.

Furthermore, the growing concept of smart cities emphasizes the need for intelligent infrastructure capable of improving public services and environmental sustainability. The Smart Dustbin Segregation System contributes to this vision by promoting cleaner surroundings, reducing operational costs, and enhancing recycling processes.

1.2 Scope of the Project

The scope of this project includes the design, development, and implementation of an IoT-Based Smart Dustbin Segregation System capable of automatically identifying and segregating waste. The system uses sensors, a microcontroller, and a servo motor to classify and direct waste into appropriate compartments.

The project covers:

- Detection of waste using sensors.
- Automatic segregation of waste into designated bins.
- Control of the segregation mechanism using a microcontroller.
- Monitoring of dustbin status through IoT technology.
- Testing and evaluation of system performance.

The proposed system is intended for small-scale applications such as homes, schools, offices, and public places. Future enhancements can include AI-based image recognition, cloud integration, mobile application support, and large-scale smart city deployment.

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Statement

Improper waste disposal and lack of segregation at the source have become significant environmental concerns. In many households, offices, and public places, different types of waste such as biodegradable and non-biodegradable materials are disposed of together. This makes recycling and waste processing difficult, increases the amount of waste sent to landfills, and contributes to environmental pollution. An efficient and automated waste segregation system is needed to improve waste management practices and support sustainable development. Manual waste segregation is often time-consuming, inefficient, and prone to human error. Lack of awareness and proper waste management systems further worsen the problem. Therefore, there is a need for a smart, automated solution that can accurately identify and segregate waste to promote cleaner surroundings and environmental sustainability.

2.2 Problem Definition

The objective of this project is to design and develop an IoT-Based Smart Dustbin Segregation System that can automatically identify and separate waste into appropriate categories. The system uses sensors and a microcontroller to detect the characteristics of the waste and control a servo motor that directs the waste into the correct compartment. This automation reduces human effort, improves segregation accuracy, and enhances the overall efficiency of waste management. The system aims to promote proper waste disposal practices and support environmental sustainability. By automatically segregating wet and dry waste, it helps simplify the recycling and waste treatment process. The project also demonstrates the practical application of IoT and embedded systems in solving real-world waste management challenges. The proposed system provides a cost-effective, reliable, and user-friendly solution for smart waste management in homes, offices, and public places.

2.3 Challenges

1. Accurate Identification of Waste Materials

Different waste materials may have similar physical characteristics, making it challenging for sensors to accurately classify them. Ensuring correct segregation is essential for the effectiveness of the system.

2. Sensor Calibration and Reliability

Sensors must be properly calibrated to provide accurate readings. Environmental factors such as dust, moisture, and lighting conditions can affect sensor performance and reliability.

3. Real-Time Operation

The system must process sensor data and operate the segregation mechanism quickly to ensure smooth and efficient waste disposal without delays.

4. Cost-Effective Implementation

Developing a smart waste segregation system using affordable components while maintaining accuracy and performance is an important challenge.

5. Power Management and Maintenance

The system requires efficient power utilization for continuous operation. Regular maintenance of sensors, motors, and electronic components is necessary to ensure long-term reliability.

CHAPTER 3

OBJECTIVES

The main objective of the IoT-Based Smart Dustbin Segregation System is to develop an intelligent waste management solution that can automatically identify and segregate different types of waste. The system uses sensors, a microcontroller, and IoT technology to reduce manual effort, improve segregation accuracy, and make waste disposal more efficient. By ensuring proper separation of waste at the source, the system supports recycling activities and helps reduce environmental pollution.

In addition, the project aims to promote smart waste management practices through automation and real-time monitoring. The IoT-enabled features allow users to track dustbin status and waste levels, enabling timely collection and preventing overflow. The system is designed to be cost-effective, user-friendly, and suitable for use in homes, schools, offices, and public places, contributing to a cleaner and more sustainable environment.

3.1 Primary Objectives

The primary objective of this project is to develop an IoT-Based Smart Dustbin Segregation System that can automatically identify and segregate waste into different categories. The system uses sensors, a microcontroller, and a servo motor to reduce manual effort and improve the efficiency of waste management.

The specific objectives are:

- To automatically detect and classify different types of waste.
- To segregate waste into appropriate compartments without human intervention.
- To improve the accuracy and efficiency of waste segregation.
- To reduce the time and labour required for manual waste sorting.
- To promote proper recycling and waste disposal practices.
- To monitor dustbin status and waste levels using IoT technology.
- To prevent overflow of waste bins through timely monitoring.
- To develop a cost-effective and user-friendly smart waste management solution.

3.2 Secondary Objectives

1. Reduce Manual Effort in Waste Segregation

To minimize the need for human involvement in sorting waste by automating the segregation process using smart technologies.

2. Improve Recycling Efficiency

To ensure that recyclable and non-recyclable waste are separated correctly, thereby enhancing the effectiveness of recycling operations.

3. Promote Smart Waste Management Practices

To encourage the adoption of intelligent waste management solutions that contribute to cleaner surroundings and sustainable development.

4. Monitor Waste Collection Using IoT Technology

To enable real-time monitoring of dustbin status and waste levels, allowing timely waste collection and preventing bin overflow.

5. Enhance Environmental Awareness

To create awareness about proper waste segregation and its importance in reducing pollution and conserving natural resources.

6. Develop a Cost-Effective Smart Solution

To design a low-cost and efficient waste segregation system that can be easily implemented in homes, schools, offices, and public places.

CHAPTER 4

SYSTEM DESIGN

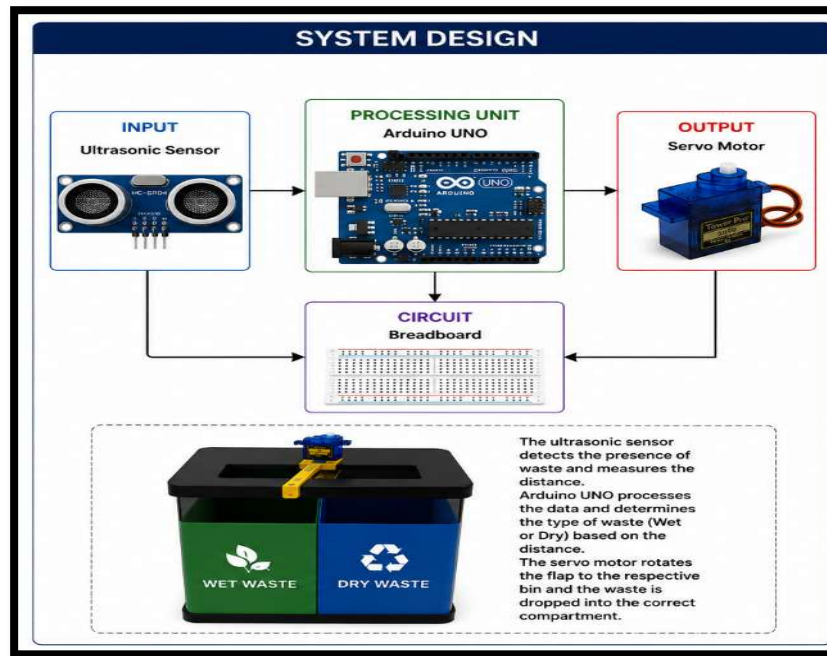


Fig 4: System Design of Smart Dustbin with Waste Segregation using IoT showing the user depositing waste, sensor-based waste detection, Arduino Uno processing, automatic segregation using a servo motor, real-time monitoring, and data storage for waste management.

User Interaction

- The user places waste into the smart dustbin for automatic segregation.

Sensor Module

- The sensors detect the waste and collect the required data for classification.

Processing Unit (Arduino Uno)

- The Arduino Uno processes sensor data and controls the waste segregation operation.

Segregation Mechanism

- The servo motor directs the waste into the appropriate wet or dry waste compartment.

4.1 System Architecture

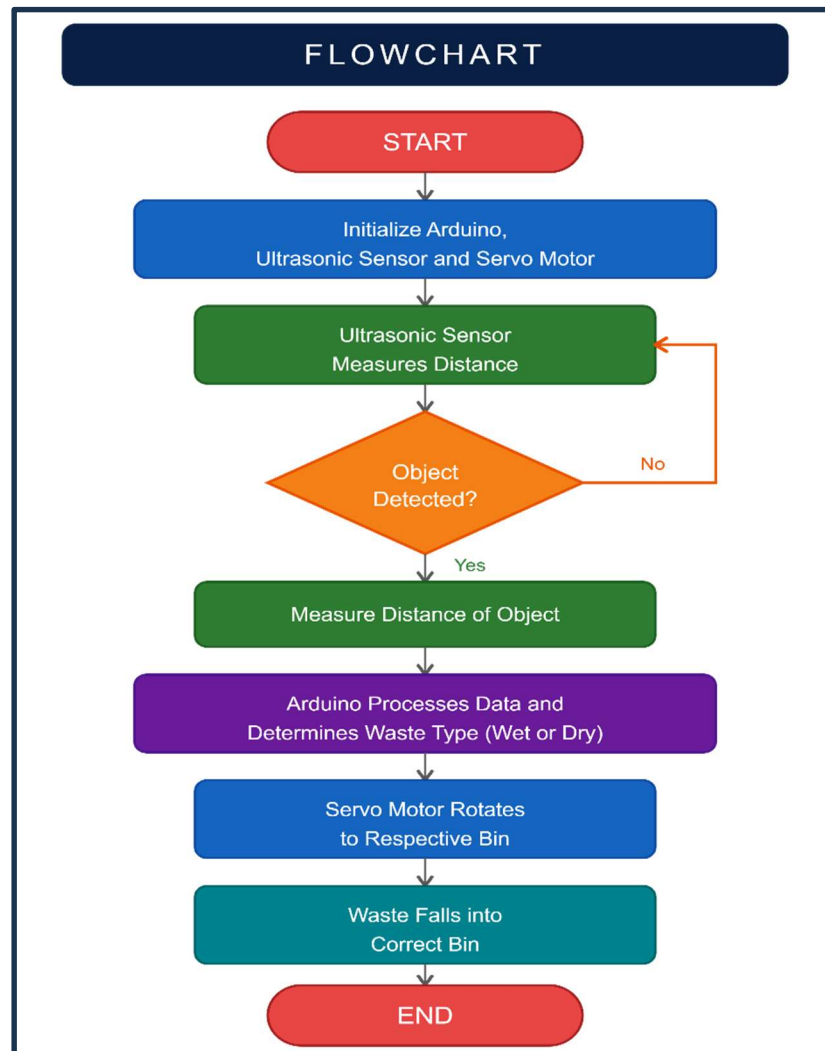


Fig 4.1.1: System Architecture of Smart Dustbin with Waste Segregation using IoT illustrating the flow of data from the Ultrasonic Sensor and Moisture Sensor to the Arduino Uno controller, which processes the sensor inputs and controls the Servo Motor for automatic waste segregation. The system also integrates an IoT monitoring module and database for waste level tracking, data storage, and remote monitoring.

4.2 Workflow

The working process of the system follows a sequence of steps:

1. The user places waste into the input section of the smart dustbin.
2. Sensors detect the waste and collect the required information.
3. The microcontroller processes the sensor data to identify the waste category.
4. Based on the classification result, the servo motor rotates to the appropriate position.
5. The waste is directed into the corresponding collection compartment.
6. The system updates the waste status and transmits the information through the IoT module.
7. Users or administrators can monitor the dustbin status in real time.

4.3 Model Selection

A sensor-based classification model is selected for this project because it provides a simple, reliable, and cost-effective solution for waste segregation. Unlike advanced machine learning models that require large datasets and high computational resources, sensor-based classification can operate efficiently using basic hardware components. It is suitable for small-scale applications such as homes, schools, offices, and public places.

Reasons for Selecting the Sensor-Based Model

- Simple design and implementation.
- Low hardware and maintenance cost.
- Faster processing and response time.
- Low power consumption.
- Easy integration with IoT technology.
- Suitable for real-time waste segregation.
- Reliable performance for small-scale

CHAPTER 5

REQUIREMENTS

5.1 Hardware Requirements

- ESP32/Arduino Uno
- Servo Motor
- Ultrasonic Sensor
- Jumper Wires
- Breadboard
- Power Supply
- Dustbin Structure

5.2 Software Requirements

- Arduino IDE
- Embedded C/C++
- IoT Monitoring Platform

5.3 Functional Requirements

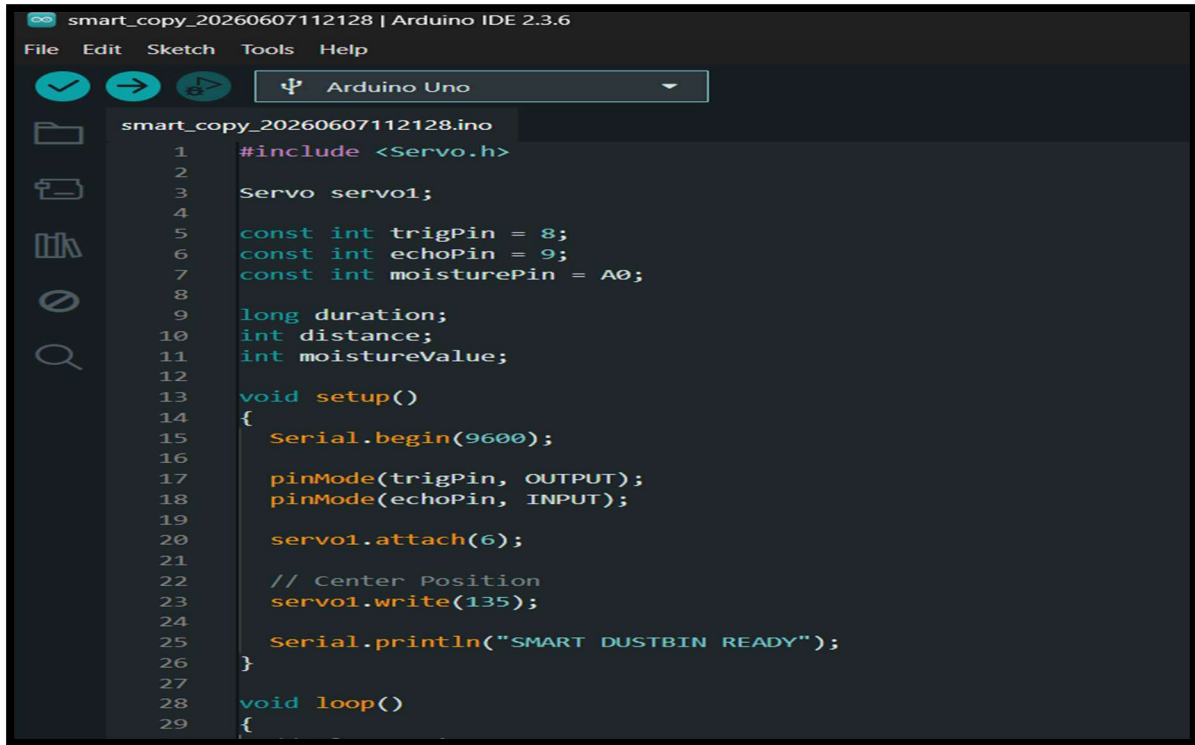
- Detect waste input.
- Classify waste category.
- Operate segregation mechanism.
- Monitor bin status.

5.4 Non-Functional Requirements

- Scalability
- Low Power Consumption
- User friendly

CHAPTER 6

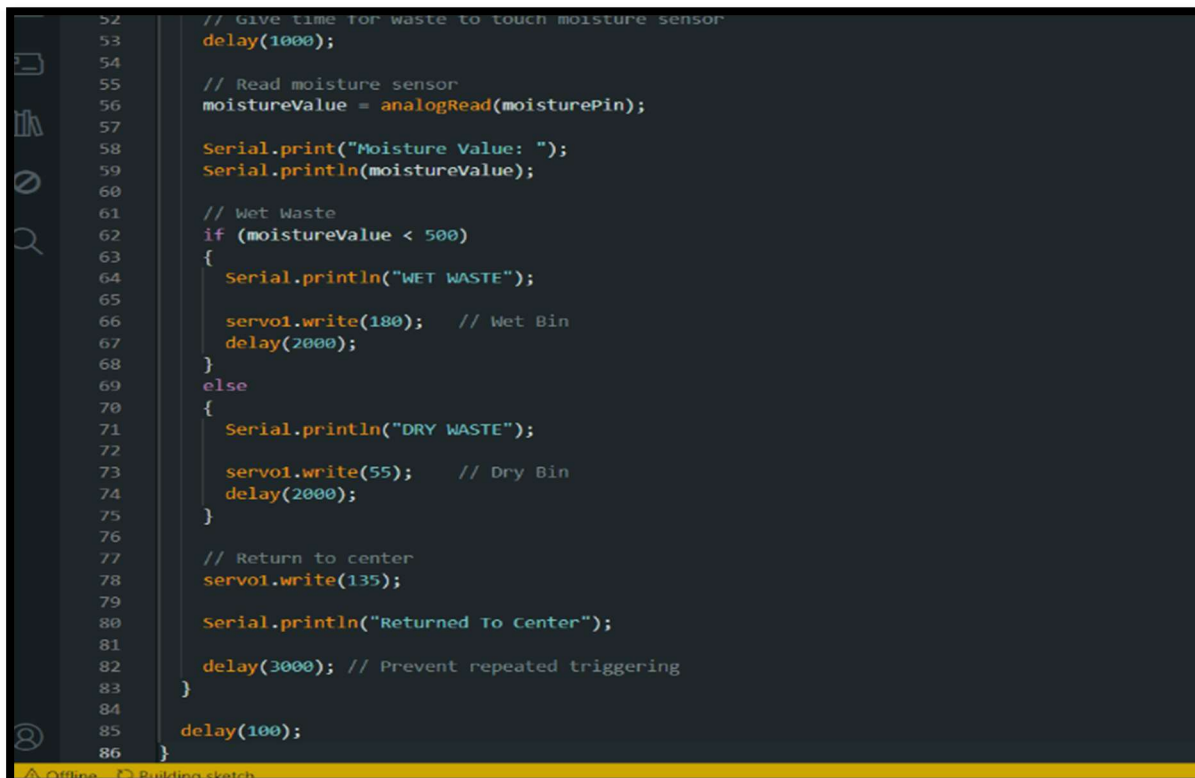
IMPLEMENTATION



```

smart_copy_20260607112128 | Arduino IDE 2.3.6
File Edit Sketch Tools Help
smart_copy_20260607112128.ino
1  #include <Servo.h>
2
3  Servo servo1;
4
5  const int trigPin = 8;
6  const int echoPin = 9;
7  const int moisturePin = A0;
8
9  long duration;
10 int distance;
11 int moistureValue;
12
13 void setup()
14 {
15   Serial.begin(9600);
16
17   pinMode(trigPin, OUTPUT);
18   pinMode(echoPin, INPUT);
19
20   servo1.attach(6);
21
22   // Center Position
23   servo1.write(135);
24
25   Serial.println("SMART DUSTBIN READY");
26 }
27
28 void loop()
29 {

```



```

52 // Give time for waste to touch moisture sensor
53 delay(1000);
54
55 // Read moisture sensor
56 moistureValue = analogRead(moisturePin);
57
58 Serial.print("Moisture Value: ");
59 Serial.println(moistureValue);
60
61 // Wet Waste
62 if (moistureValue < 500)
63 {
64   Serial.println("WET WASTE");
65
66   servo1.write(180); // Wet Bin
67   delay(2000);
68 }
69 else
70 {
71   Serial.println("DRY WASTE");
72
73   servo1.write(55); // Dry Bin
74   delay(2000);
75 }
76
77 // Return to center
78 servo1.write(135);
79
80 Serial.println("Returned To Center");
81
82 delay(3000); // Prevent repeated triggering
83 }
84
85 delay(100);
86 }

```



```

30 // Ultrasonic Sensor
31 digitalWrite(trigPin, LOW);
32 delayMicroseconds(2);
33
34 digitalWrite(trigPin, HIGH);
35 delayMicroseconds(10);
36
37 digitalWrite(trigPin, LOW);
38
39 duration = pulseIn(echoPin, HIGH);
40
41 distance = duration * 0.034 / 2;
42
43 Serial.print("Distance: ");
44 Serial.print(distance);
45 Serial.println(" cm");
46
47 // Waste detected
48 if (distance > 0 && distance < 20)
49 {
50     Serial.println("Waste Detected");
51
52     // Give time for waste to touch moisture sensor
53     delay(1000);
54
55     // Read moisture sensor
56     moistureValue = analogRead(moisturePin);
57
58     Serial.print("Moisture Value: ");
59     Serial.println(moistureValue);
60
61     // Wet Waste
62     if (moistureValue < 500)
63     {

```

Fig 6: Arduino source code of the Smart Dustbin with Waste Segregation system illustrating ultrasonic sensor initialization, distance measurement, waste detection, moisture value analysis, wet/dry waste classification, and servo motor actuation for directing waste into the appropriate bin.

6.1 Project Structure Overview

The implementation of the IoT-Based Smart Dustbin Segregation System combines hardware and software components to achieve automatic waste segregation and monitoring. The system consists of sensors for waste detection, a microcontroller for processing data, a servo motor for directing waste into appropriate compartments, and an IoT module for monitoring and communication. All components work together to provide an efficient and automated waste management solution.

6.2 Key Components

1. Sensor Module

The sensor module is responsible for detecting the presence and characteristics of waste. It collects data and sends it to the microcontroller for processing.

2. Processing Unit

The processing unit consists of the Arduino/ESP32 microcontroller. It acts as the brain of the system by receiving sensor data, analyzing it, and making decisions regarding waste classification and segregation.

3. Motor Control Unit

The motor control unit manages the operation of the servo motor. Based on instructions from the microcontroller, the servo motor rotates to the required position and directs waste into the correct compartment.

4. Waste Collection Bins

Separate bins are used for collecting different categories of waste. Proper segregation at this stage helps improve recycling efficiency and waste management practices.

5. IoT Monitoring Module

The IoT module enables real-time monitoring of the dustbin status. It transmits information such as waste levels, bin capacity, and system status to users or administrators for effective management.

6.3 Frontend (Monitoring Interface)

The frontend serves as the user interface of the system. It displays important information collected from the smart dustbin and allows users to monitor its status easily.

Functions of the Frontend

- Displays current waste levels in the bins.
- Shows system status and operational information.
- Provides alerts when bins are nearly full.
- Enables real-time monitoring through IoT connectivity.
- Presents data in a user-friendly format.

6.4 Backend

The backend is responsible for processing sensor data and controlling the overall operation of the system. It acts as the communication layer between the hardware components and the monitoring interface.

Functions of the Backend

- Receives data from sensors.
- Processes and analyzes sensor readings.
- Determines the waste category.
- Controls the servo motor for segregation.
- Sends updates to the IoT monitoring platform.
- Maintains system reliability and performance.

6.5 Data Processing and Storage

The system continuously collects information from sensors and processes it through the microcontroller. The processed data is then stored and transmitted for monitoring purposes.

Data Processing Steps

1. Sensor data is collected from the waste detection module.
2. The microcontroller analyzes the received data.
3. The waste type is identified.
4. The segregation mechanism is activated.
5. Data is transmitted through the IoT module.

Data Storage Benefits

- Maintains records of waste levels.
- Supports monitoring and analysis.
- Helps optimize waste collection schedules.
- Enables future system improvements based on collected data.

The implementation successfully integrates automation and IoT technologies to create an efficient smart waste management system that improves waste segregation accuracy and reduces manual effort.

CHAPTER 7

RESULTS AND EVALUATION

The Smart Dustbin with Waste Segregation using IoT system was successfully designed and implemented. The developed system effectively detects waste using an ultrasonic sensor, identifies the type of waste using a moisture sensor, and automatically segregates it into wet and dry waste categories using a servo motor.

During testing, the ultrasonic sensor accurately detected the presence of waste within the predefined distance range. Once waste was detected, the moisture sensor measured its moisture content and provided data to the Arduino Uno microcontroller. Based on the sensor readings, the system correctly classified the waste as either wet or dry and directed it to the appropriate compartment.

The servo motor responded efficiently to the classification results and successfully rotated to the required position for waste segregation. After disposing of the waste, the servo motor returned to its default position, preparing the system for the next operation.

The system demonstrated reliable performance in automating waste segregation, reducing manual intervention, and improving waste management efficiency. The results indicate that the proposed solution can contribute to cleaner environments, better waste handling practices, and increased awareness of smart waste management technologies.

Overall, the Smart Dustbin achieved its intended objectives by providing an effective, low-cost, and environmentally friendly approach to automatic waste segregation and monitoring

7.1 User Interface Snapshots

Fig 7.1.1 The Smart Dustbin with Waste Segregation using IoT was successfully developed using an Arduino Uno, Ultrasonic Sensor, Moisture Sensor, and Servo Motor. The system automatically detects waste, identifies whether it is wet or dry, and directs it into the appropriate compartment.

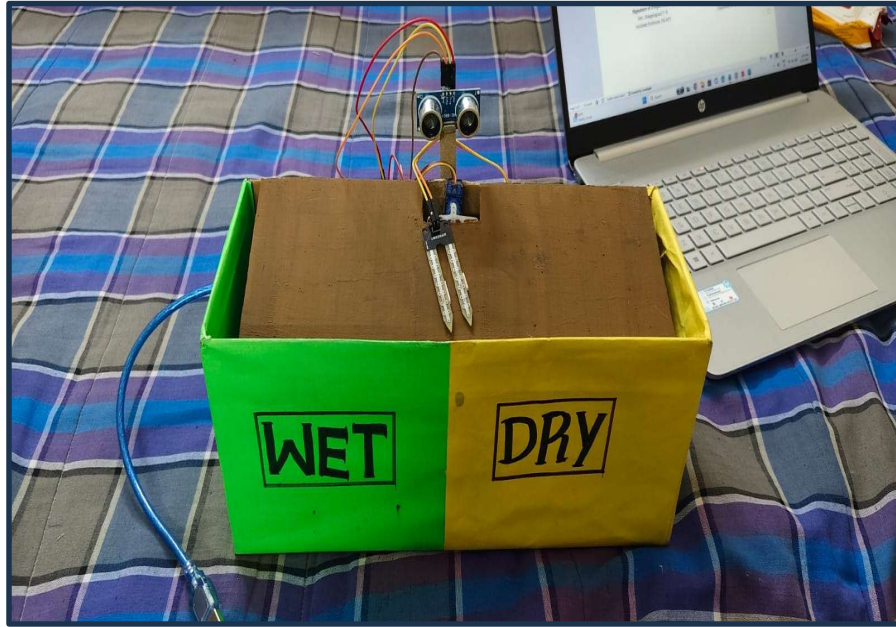
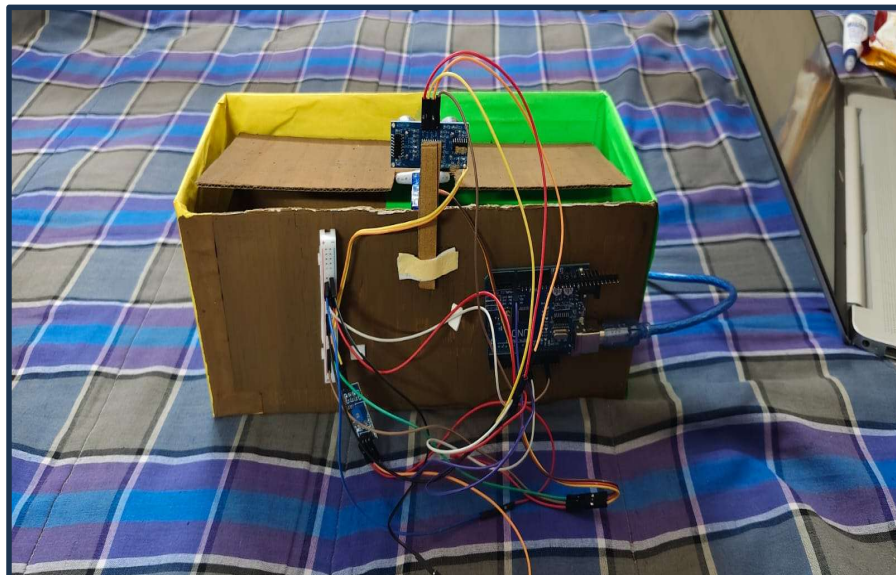


Fig 7.1.2 During testing, the ultrasonic sensor accurately detected the presence of waste, while the moisture sensor effectively classified the waste based on its moisture content. The servo motor responded correctly by rotating the divider towards the wet or dry section. The prototype demonstrated reliable performance and successful waste segregation in most test cases.



CHAPTER 8

CONCLUSION

8.1 Summary of Work

The IoT-Based Smart Dustbin Segregation System was successfully developed to address the problem of improper waste segregation. The project integrated sensors, a microcontroller, a servo motor, and IoT technology to create an automated waste management solution. The system was capable of detecting waste, classifying it, and directing it into the appropriate compartment without human intervention.

The implementation demonstrated the effectiveness of automation in improving waste management practices. Real-time monitoring capabilities further enhanced the functionality of the system by providing information about dustbin status and waste levels. Overall, the project achieved its objectives of reducing manual effort, improving waste segregation efficiency, and promoting environmentally responsible waste disposal.

8.2 Future Improvements

1. AI-Based Image Recognition

Machine learning and image processing techniques can be incorporated to identify waste materials more accurately.

2. Mobile Application Integration

A dedicated mobile application can be developed for real-time monitoring, notifications, and system management.

3. Additional Waste Categories

The system can be extended to classify and segregate more waste categories such as plastic, metal, paper, glass, and electronic waste.

4. Solar-Powered Operation

The system can be powered using solar energy to reduce electricity consumption and improve sustainability.



YouTube Link : <https://youtu.be/WaROA0SYDII?si=UZjeItcJ0a3ldeV9>